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Formalizing Domain Theory in the Age of AI

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11th International Symposium on Domain Theory and Its Applications
Chengdu, China
6 August 2026

Outline

1. **A challenge of mathematical practice**
2. **AI4Math at the research frontier**
3. **A possible AI-era workflow**
4. **Verification with Lean 4**
5. **VeriLoop: an LLM + Lean verification loop**
6. **Formalizing a classic Domain Theory textbook**

Terence Tao: AI is creating a new crisis for mathematics

- **A crisis of scale and trust.** AI can generate plausible proofs much faster than humans can check them. An AI-generated proof may sound convincing but still hide an error or a changed assumption.
- **A crisis of attention.** When proofs become abundant, which results are meaningful and worth attention?
- **A crisis of identity.** AI may surpass humans in proving theorems, writing papers, and conducting peer reviews. What roles remain for human mathematicians, academic institutions, and journals?

Not an inconsistency crisis inside mathematics;
a new crisis of mathematical values and practices.

Source: T. Tao, *Mathematics in the age of AI*, ICM 2026, esp. slides 4 and 43–46.

Recent breakthroughs by AI

Planar unit-distance conjecture

Problem Asymptotic unit-distance bound.
Age 80 years (1946–2026).
Date 20 May 2026 (public proof).
Tool OpenAI reasoning model.
Status Expert-checked; no Lean.

Jacobian conjecture in dimension 3

Problem Constant-Jacobian invertibility in complex 3-space.
Age 87 years (1939–2026).
Date 20 July 2026 (announced).
Tool Claude Fable + L. Alpöge.
Status Formally checked; unrefereed.

Six Erdős problems

Problem #390, 486, 536, 788, #1002, 1038.
Age 46–68 years.
Date 24 July 2026 (candidates).
Tool GPT-5.6 Sol + Codex.
Status Candidates; review incomplete.

Dinitz–Garg–Goemans conjecture

Problem Cost-preserving flow rounding.
Age 27 years (1999–2026).
Date 22 July 2026 (certificate).
Tool GPT-5.6 Pro + D. Rybin.
Status Certificate; review pending.

Sources: OpenAI unit-distance report; Wang's Erdős-problem report; AFP Jacobian entry; Rybin's Dinitz–Garg–Goemans report. Dates are announcement dates.

Frontier AI systems in mathematics

- **ChatGPT 5.6 Pro — IMO 2026.** It solved all six problems, surpassing the best human score.
- **First Proof / UCLA Moonshot Harness.** On ten unpublished research problems, four systems collectively solved **7 of 10 problems** under expert review.
- **DeepSeekMath-V2 + DeepSeek-Prover-V2.** These systems combine self-verification for long-form proofs with recursive subgoal decomposition and Lean 4 proof synthesis.
- **PKU AI4Math: Rethlas + Archon (Bin Dong's team).** Rethlas found a counterexample to the 2014 Anderson conjecture; Archon produced an approximately 19,000-line Lean 4 verification with essentially no human involvement.
- **Google's AI co-mathematician.** It provides a stateful research workbench spanning ideation, literature search, computation, theorem proving, and theory building.

Sources: AutoFyn; First Proof; DeepSeekMath-V2; DeepSeek-Prover-V2; Rethlas/Archon; Google AI co-mathematician.

OpenAI's ten open-problem advances

10 open-problem advances

Released together by OpenAI on 1 August 2026

- OpenAI reports ten advances on mathematical problems over different areas whose main results had seen no progress for at least a decade—and usually much longer.

- **Generation and formal verification arrived together.** OpenAI's internal Astra generated the arguments; humans used it to prepare manuscripts; every argument received a Lean certificate.

Another "AlphaGo moment"?

AI vs. **humans**

Sources: OpenAI announcement; 249-page collection; Lean certificates.

Our focus

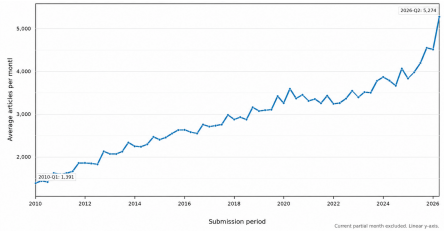
Rather than attempting to tell the whole story of how AI will change mathematics, this talk focuses only on one concrete shift.

AI is making candidate proofs increasingly abundant, while verification, understanding, and knowledge integration remain scarce.

candidate volume
rising fast
>
verification capacity
still human-scale

This shift is already under way.

Mathematics submissions to arXiv are rising



Average monthly submissions by calendar quarter; 2026 Q2 reached 5,274.

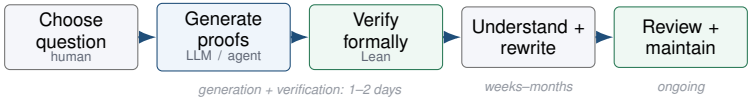
Data: arXiv Monthly Submissions; chart aggregation: J. Dekoninck, 5 July 2026. The chart does not establish AI as the cause.

A possible future mathematical workflow—if we are lucky

Traditional workflow



AI-era workflow



Classical bottleneck
Discovering and writing a plausible proof.

New bottleneck
Verifying, understanding, and integrating proofs.

Timescales are illustrative, problem-dependent examples, not empirical averages.

Automatic verification becomes necessary

Human review cannot scale linearly with model output. **Automatic verification can protect our attention.**

How can we automate verification?

Lean 4

open-source

- an **interactive theorem prover**,
- a **functional programming language**,
- based on **dependent type theory**.

Faithful translation + successful kernel check $\Longrightarrow$ the encoded proof is correct.

We also need LLMs to translate proofs into Lean code.

Source: Lean Language Reference, Introduction and "Validating a Lean Proof."

Lean 4 is based on type theory

proposition $P : \text{Prop} \implies$ **proof term** $p : P$

A mathematical example: monotone maps compose

If $f : A \to B$ and $g : B \to C$ preserve order, then $g \circ f$ also preserves order.

```
example {A B C : Type*}
    [Preorder A] [Preorder B] [Preorder C]
    (f : A -> B) (g : B -> C)
    (hf : Monotone f) (hg : Monotone g) :
    Monotone (fun x => g (f x)) := by
  intro x y hxy
  have h_after_f : f x <= f y := by
    apply hf
    exact hxy
  have h_after_g : g (f x) <= g (f y) := by
    apply hg
    exact h_after_f
  exact h_after_g
```

$$x \leq y \implies f(x) \leq f(y) \implies g(f(x)) \leq g(f(y))$$

This example was compiled with Lean 4 and mathlib.

Scott continuity: order theory meets topology

Order-theoretic definition

Monotone; preserves existing directed suprema.

```
def ScottContinuous (f : A -> B) : Prop :=
  And (Monotone f)
    (forall {S : Set A} {x : A},
      DirectedSet S -> IsSup S x ->
        IsSup (Set.image f S) (f x))
```

Scott-topological definition

Preimages of Scott-open sets remain Scott-open.

```
def TopologicallyScottContinuous
    (f : A -> B) : Prop :=
  forall O : Set B, ScottOpen O ->
    ScottOpen (Set.preimage f O)
```

The classical theorem

```
example {f : A -> B} :
    Iff (ScottContinuous f)
        (TopologicallyScottContinuous f) :=
  prop_2_3_4_scottContinuous_iff_topological
```

$\Rightarrow$ Directed-supremum preservation makes inverse images Scott-open.

$\Leftarrow$ Topological continuity recovers monotonicity and supremum preservation.

Source theorem: Abramsky–Jung, *Domain Theory*, Proposition 2.3.4; Lean development: DomainTheory_Lean4_texbook.

Inside Proposition 2.3.4: the actual Lean source

$\Rightarrow$

```
theorem prop_2_3_4_forward
    {β : Type v} [Preorder β]
    {f : α → β}
    (hf : ScottContinuous f) :
    TopologicallyScottContinuous f := by
  intro O hO
  constructor
  • intro x y hy hxy
    exact hO.1 hy (hf.1 hxy)
  • intro A s hdir hA hs
    have hsImg :
        IsSup (f '' A) (f s) :=
      hf.2 hdir hs
    exact hO.2
      (DirectedSet.image hf.1 hdir)
      (by
        intro z hz
        rcases hz with
          ⟨a, ha, rfl⟩
        exact hA ha)
      hsImg
```

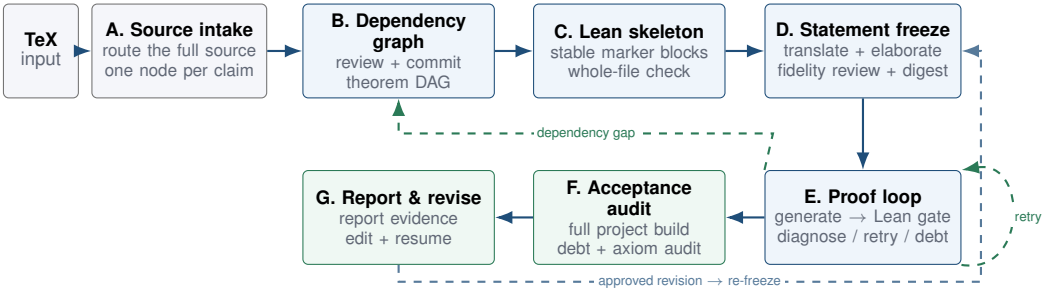
Complete equivalence theorem

```
theorem
    prop_2_3_4_scottContinuous_iff_topological
    {β : Type v} [Preorder β] {f : α → β} :
    ScottContinuous f ↔
      TopologicallyScottContinuous f := by
  constructor
  • exact prop_2_3_4_forward
  • intro hf
    have hmono : Monotone f := by
      intro x y hxy
      apply prop_2_3_2_open_to_order
      intro O hO hfxO
      exact prop_2_3_2_order_to_open hxy
        (f ⁻¹' O) (hf O hO) hfxO
    constructor
    • exact hmono
    • intro A x hA hx
      constructor
      • intro y hy
        rcases hy with ⟨a, ha, rfl⟩
        exact hmono (hx.1 ha)
      • intro y hy
        apply prop_2_3_2_open_to_order
        intro O hO hfxO
        rcases
            (prop_2_3_3_sup_iff_neighborhoods
              hA).1 hx |>.2
            (f ⁻¹' O) ⟨hf O hO, hfxO⟩
          with ⟨a, ha, hfaO⟩
        exact scottOpen_upper hO hfaO
          (hy ⟨a, ha, rfl⟩)
```

Exact excerpts: ScottTopology.lean and Chapter02.lean.

Our VeriLoop: a governed TeX-to-Lean verification loop

Models propose; **independent review and deterministic Lean gates decide**.



Workflow source: TexLean Loop 0.3.0, `README.zh-CN.md`, `DESIGN_PRINCIPLES.md`, and `docs/LOOP_DESIGN.md`.

One workflow example: a counterexample in domain theory

The problem

Xu Problems 1.14–1.15

If P is quasicontinuous and nonquasialgebraic, must

1. $[0,1]$ be a Scott-continuous retract of P?
2. P contain a sub-dcpo order-isomorphic to $[0,1]$?

More open problem examples:
https://chanyuxu.github.io/ai4math/

The candidate counterexample

Basic dyadic intervals $n \geq 1,\ 0 \leq k \leq 2^n$

$$J_{n,k} = X \cap \left[\frac{k-1}{2^n},\ \frac{k+1}{2^n}\right].$$

$$\mathcal{B} = \left\{ I = \bigcap_{i=1}^{r} J_{n_i,k_i} : r \geq 1,\ \operatorname{diam}(I) > 0 \right\}.$$

$$P = \{\varnothing\} \cup \{\{x\} : x \in X\} \cup \mathcal{B},$$

$$A \leq_P B \iff B \subseteq A.$$

- $\mathcal{B}$ is countable; every member is a nondegenerate closed interval in $X = [0,1]$.
- P is a quasicontinuous complete lattice, but is neither continuous nor quasialgebraic.
- Every chain in P is countable.

Why this is a counterexample

Every chain in P is countable, whereas $[0,1]$ is an uncountable chain. Thus, there is no order embedding $X \hookrightarrow P$, and hence no Scott-continuous retraction of P onto X.

Problem source: X. Xu, "Two problems related to quasicontinuous domains," pp. 30–34

What the workflow produced: an auditable Lean project

17	3,809	0	build passed
Lean modules	lines of Lean	local proof debt	focused target

Generated module structure

Foundations and lattice (5)

`Basic` · `SparseInterval` · `IntervalGeometry` · `FiniteGrid` · `Lattice`

Approximation machinery (6)

`Neighborhoods` · `SelectedNeighborhoods` · `SkeletonFiniteness` · `AbstractApproximation` · `ConcreteApproximation` · `SourceFamilies`

Main results (5)

`Quasicontinuity` · `Noncontinuity` · `Counterexample` · `Splitting` · `Main`

Public facade (1)

`UnitIntervalRetracts.lean`

The main theorem depends on this substantial, modular development.

Supporting theorem: `isQuasicontinuous_P`

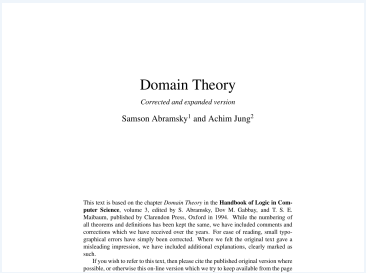
Finite approximants are filtered, and their intersection is $\uparrow K$.

```
theorem isQuasicontinuous_P : IsQuasicontinuous P := by
  intro K
  change FilteredUnderInclusion (quasiUpperFamily K) ∧
    ⋂₀ (quasiUpperFamily K) =
      AbramskyJung.Chapter2.upperClosure ({K} : Set P)
  exact ⟨quasiUpperFamily_filtered K,
    sInter_quasiUpperFamily K⟩
```

The main theorem: `mainCounterexample`

```
theorem mainCounterexample : MainCounterexampleProperties where
  dcpo := P_isDcpoOrder
  quasicontinuous := isQuasicontinuous_P
  notContinuous := not_isContinuousDomain_P
  notQuasialgebraic := not_isQuasialgebraic_P
  chainsCountable := fun {_C} hC => chain_countable hC
  quotientScottContinuous := q_scottContinuous_surjective.1
  quotientSurjective := q_scottContinuous_surjective.2
  noIntervalOrderEmbedding := no_orderEmbedding_X_to_P
  noIntervalRetract := no_scottContinuous_retract_X_of_P
```

A project to translate a classic domain theory text into Lean



Domain Theory

Corrected and expanded version

Samson Abramsky[1] and Achim Jung[2]

This text is based on the chapter *Domain Theory* in the **Handbook of Logic in Computer Science**, volume 3, edited by S. Abramsky, Dov M. Gabbay, and T. S. E. Maibaum, published by Clarendon Press, Oxford in 1994. While the numbering of all theorems and definitions has been kept the same, we have included comments and corrections which we have received over the years. For ease of reading, small typographical errors have simply been corrected. Where we felt the original text gave a misleading impression, we have included additional explanations, clearly marked as such.

If you wish to refer to this text, then please cite the published original version where possible, or otherwise this on-line version which we try to keep available from the page

Abramsky & Jung, *Domain Theory*

The book

Samson Abramsky and Achim Jung's *Domain Theory*, the corrected and expanded version of their 1994 chapter in the *Handbook of Logic in Computer Science*.

Expected outcome: a reusable library

- **Faithful.** Every theorem remains traceable to the source, with its hypotheses and conventions reviewed.
- **Complete.** Proof boundaries are explicit: proved, admitted, axiomatized, blocked, or under review.
- **Reusable.** Definitions and lemmas form a stable, mathlib-oriented API for later chapters and research.
- **Interactive.** Participants can collaborate, review, and edit.

Source: S. Abramsky and A. Jung, *Domain Theory*, corrected and expanded version of the 1994 handbook chapter.

Progress: Chapter 2

Source alignment

Source items mapped	**57**
Missing or ambiguous declarations	**0**
Reviewed TeX overrides	**57 / 57**

Verification status

Focused Chapter 2 build	**passes** with warnings
Explicit local axioms	**3**
Book-omitted proofs (`admit`)	**2**

What the numbers mean. All 57 mapped items have reviewed TeX overrides, while three local axioms and two book-omitted proofs remain explicit. This is progress, not project completion.

Source: DomainTheory_Lean4_texbook; Chapter 2 map/review audit and focused build, 30 July 2026.

Join us in building verified Domain Theory

Why this project matters

- **Provide infrastructure for human–AI research.** A trusted library gives mathematicians and AI systems reusable definitions, lemmas, and kernel-checked feedback.
- **Make verification cumulative.** Every checked result becomes a dependable foundation for future proofs.
- **Accelerate LLM translation with high-quality data.** Source-aligned textbook–Lean pairs support iterative improvement and evaluation.

Long-term goal. Contribute Lean formalizations of classical Domain Theory results to **mathlib**.

We welcome domain theorists, Lean contributors, AI-for-math researchers, and students to build this library together.

Project URL

https://github.com/ChanYuxu/domain-theory-book-formalization

A source-aligned interface for reviewing the formalization

DT **Domain Theory**
Lean 4 formalization

CHAPTER 2 · Domains individually

Read-only leanprover/lean4:v4.30.0-r… GitHub

Find an item

15 pages · 57/57 markers

- **Definition 2.1.1** Preorders and partial orders
- **Proposition 2.1.2** Order duals
- **Definition 2.1.3** Order-theoretic vocabulary
- **Proposition 2.1.4** Elementary laws for suprema and infima
- **Proposition 2.1.5** Closure systems as complete lattices
- **Definition 2.1.6** Monotone maps
- **Proposition 2.1.7** Fixed points in complete lattices
- **Definition 2.1.8** Directed sets, ideals, filters, and chains

1

Domains individually

Lean 4 Formalisation Companion – Chapter 2

This project-generated reading copy contains recovered mathematical items, short source locators, and redrawn diagrams. It does not reproduce the original PDF or automatic OCR prose. Click a Lean tag in the web reader to open the mapped Lean declaration.

Section 2.1

Source PDF page 10

Definition 2.1.1. A set P with a binary relation $\sqsubseteq$ is called a partially ordered set or poset if the following holds for all $x, y, z \in P$:

1. $x \sqsubseteq x$ (Reflexivity)
2. $x \sqsubseteq y \wedge y \sqsubseteq z \Rightarrow x \sqsubseteq z$ (Transitivity)
3. $x \sqsubseteq y \wedge y \sqsubseteq x \Rightarrow x = y$ (Antisymmetry)

If antisymmetry is omitted, the structure is a preorder. The induced equivalence relation is $\sqsubseteq \cap \sqsupseteq$, and passing to equivalence classes gives a canonical poset. A common way to extract order-theoretic content from a relation R is its transitive closure $\bigcup_{n \in \mathbb{N}^+} R^n$.

Lean

Proposition 2.1.2. If $(P, \sqsubseteq)$ is an ordered set, then so is $P^{\mathrm{op}} = (P, \sqsupseteq)$.

Lean



The flat booleans The four-element lattice The four-element chain

Figure 1: A few posets drawn as line diagrams.



Figure 2: Three versions of the natural numbers.

Source PDF page 11

Definition 2.1.3. Let $(P, \sqsubseteq)$ be an ordered set.

Lean

1. A subset A of P is an upper set if $x \in A$ and $y \sqsupseteq x$ imply $y \in A$. We write $\uparrow A$ for the set of elements above some element of A, and abbreviate $\uparrow \{x\}$ as $\uparrow x$. The dual notions are lower set and $\downarrow A$.

1

DEFINITION 2.1.1

Preorders and partial orders

Predicate-level encodings of reflexive/transitive relations and antisymmetric partial orders.

Definition · Section 2.1 · Source PDF p. 10

DECLARATION

IsPreorderRel

Practice/DomainTheory/Chapter02.lean:41–44

Source Copy 解释

```
/-- Definition 2.1.1: a binary relation is a preorder. -/
def IsPreorderRel (r : α → α → Prop) : Prop :=
  (∀ x : α, r x x) ∧ ∀ ⦃x y z : α⦄, r x y → r y z → r x z
```

References and provenance

AI and mathematical workflow

- Tao, *Mathematics in the age of AI*, ICM 2026.
- AutoFyn, a public ChatGPT 5.6 Pro run on the IMO 2026 problems.
- First Proof Second Batch, arXiv:2606.18119.
- DeepMind, Gemini Deep Think's officially graded performance at IMO 2025.
- OpenAI, unit-distance conjecture report and proof.
- OpenAI, ten advances in mathematics and theoretical computer science; paper; Lean certificates.
- AFP record and formal-proof archive for the Alpöge–Fable counterexample.
- ITPEval: formal translation across Lean 4, Rocq, Isabelle, and HOL Light.

Formalization and project evidence

- Lean official site and documentation.
- mathlib4 repository.
- Abramsky and Jung, *Domain Theory*, corrected and expanded version.
- Current DomainTheory_Lean4_texbook checkout.
- UnitIntervalRetracts module: focused build and debt audit, 4 August 2026.
- Chapter 2 audit and focused build, 30 July 2026.

A Tale of Two Futures for Mathematics

"It was the best of times, it was the worst of times."

Charles Dickens, *A Tale of Two Cities* (1859)

THE BEST OF TIMES

Mathematicians amplified

- AI expands exploration and proof construction.
- Humans choose questions, specify meaning, and build theories.

THE WORST OF TIMES

Mathematicians displaced

- AI handles conjecture, proof, formalization, and exposition.
- Output outruns human attention; participation becomes optional.

Hilbert's Dream

Hilbert pursued formalization and finitary consistency proofs; the original program could not be completed as envisioned.

AI offers a new vision: LLMs propose proofs; Lean checks them.

Sources: C. Dickens, *A Tale of Two Cities* (1859); Stanford Encyclopedia of Philosophy, "Hilbert's Program."

Thank You

Questions & Discussion

Acknowledgment

Large language models assisted with drafting, organizing, and generating these slides.

The author remains responsible for the mathematical claims, evidence boundaries, and final presentation.